A Comparative Investigation of Broadcast Frameworks, Service Availability, and Time Transfer Performance in PPP-B2b, HAS, and MADOCA-PPP

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The precise point positioning (PPP) technique has been used in the calculations of international atomic time (TAI) links for decades. PPP depends on external precise satellite orbit, clock offset, and bias products to correct satellite-specific errors. In recent years, the emergence of real-time precise satellite products broadcast via satellites has facilitated research and applications of PPP in positioning and time transfer. These precise satellite products are provided through the PPP services of navigation satellite systems, including the BeiDou global navigation satellite system (BDS) PPP-B2b, the European Galileo high accuracy service (HAS), and the Quasi-Zenith Satellite System (QZSS) Multi-GNSS advanced orbit and clock augmentation-PPP (MADOCA-PPP). In contrast to real-time precise satellite products provided over the Internet, the bandwidth limitations of satellite broadcasts necessitate the segmentation of complete precise products into various message types for sequential transmission. Users must accurately match and reconstruct these message segments to reliably restore the complete product, thereby ensuring the continuity of PPP time transfer. Moreover, the coverage and availability of PPP supported by these satellite products directly influence the effectiveness of time transfer. Consequently, a comparative investigation of broadcast frameworks, service availability, and time transfer performance was conducted to assist time users in selecting the suitable services and addressing the challenges associated with precise product recovery. The investigation evaluated the accuracy of the recently released MADOCA-PPP for time transfer. The analysis indicates that the PPP-B2b service employs shorter message lengths and incorporates error correction mechanisms to ensure the accuracy of product transmission. In contrast, the MADOCA-PPP service provides a more straightforward message matching process, while the HAS service accomplishes global coverage through a complex encoding and transmission scheme. However, this complexity increases the challenges associated with product recovery for users. Moreover, the PPP availability of the PPP-B2b service for BDS users in China and its surrounding areas is recorded at 99.93%. In comparison, the HAS service demonstrates a comparable PPP availability of approximately 99% for both GPS and Galileo users. Furthermore, the MADOCA-PPP service achieves an impressive PPP availability of 99.99% for GPS users. In the time transfer results, where the National Time Service Center (NTSC) serves as the central node linking international GNSS service (IGS) stations, time comparison errors for all three services fluctuate within ±0.5 ns, even with a baseline length extending up to 2631 km. Notably, the newly released MADOCA-PPP demonstrates the highest time transfer accuracy, attributed to its high-precision satellite clock offset products. As a result, time users can select appropriate services to support continuous, stable, and reliable real-time time transfer at the sub-nanosecond level, based on the local continuity performance of the three services and with consideration of their broadcast frameworks, decoding complexity, and respective advantages and limitations.

Development and Broadcast of High-Precision GNSS Satellite Products

The global navigation satellite system (GNSS) is essential for positioning, navigation, and timing (PNT) applications. GNSS carrier phase time transfer using precise point positioning (PPP) [1] has been incorporated into the calculations of international atomic time (TAI) links by the International Bureau of Weights and Measures (BIPM), owing to the exceptionally low measurement noise associated with carrier phase observations [2]. The PPP technique, in addition to employing observations from a GNSS receiver, depends on externally provided high-precision satellite products. Initially, real-time satellite products were generated by the international GNSS service (IGS) analysis centers and made accessible via the Internet [3]. The emergence of real-time precise satellite products has advanced both research and applications in real-time PPP time transfer, achieving accuracy at the sub-nanosecond level [4]–[6].

In recent years, navigation systems have begun broadcasting precise satellite products directly via their satellites to address the challenges posed by limited network availability in PPP services. These include the Quasi-Zenith Satellite System (QZSS) centimeter level augmentation

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service (CLAS) [7], the BeiDou global navigation satellite system (BDS) PPP-B2b [8], the European Galileo high accuracy service (HAS) [9], and the QZSS Multi-GNSS advanced orbit and clock augmentation-PPP (MADOCA-PPP) [10]. In particular, the Galileo HAS launched its initial global services in January 2023, while the QZSS MADOCA-PPP began its operational phase in April 2024. The accuracy of the satellite-based precise products has been thoroughly analyzed and is sufficient to meet the requirements for real-time PPP [11],[12], and these products have already been applied in precise positioning. Concurrently, research on real-time PPP time transfer using satellite-based products, such as PPP-B2b and HAS products, is steadily advancing [11],[13]. However, real-time PPP time transfer imposes stringent requirements not only on the accuracy of satellite products but also on their continuous and complete reception.

In contrast to real-time precise satellite products provided via the Internet, satellite broadcasts are limited by bandwidth, necessitating the segmentation of the complete precise products into various message types for sequential transmission. Users must match and reconstruct these message segments to restore the complete product. The encoding and dissemination schemes of precise products in these three services differ, each presenting distinct advantages and limitations. Moreover, the PPP coverage and availability of these products play a critical role in ensuring the stability and reliability of PPP time transfer [11]. Notably, with the MADOCA-PPP service entering its operational phase in April 2024, there is a lack of literature regarding its application in PPP time transfer. Consequently, a comprehensive comparative analysis can assist time users in selecting the suitable service for their specific requirements while addressing any challenges that may arise during implementation.

In our contribution, we conduct an in-depth comparison of the encoding and broadcasting processes of BDS PPP-B2b, Galileo HAS, and QZSS MADOCA-PPP, focusing on the advantages of each service and the critical factors to consider during product recovery on the user side. Subsequently, we analyze PPP continuity from the users' perspective, as continuity is a key determinant of the stability and reliability of PPP time transfer. Lastly, we present the results of PPP time transfer for all three services, with particular emphasis on QZSS MADOCA-PPP, for which specific time transfer results have not been documented in the existing literature. A brief introduction to the modeling methods and processing strategies for PPP time transfer is provided, followed by a detailed comparison of the broadcast frameworks of the three services. We analyze coverage and availability, present the results of PPP time transfer, and conclude with a summary of our findings.

Method

The basic observation equations of PPP and its primary errors are initially presented, followed by an introduction to the PPP time transfer model.

PPP Model

The PPP technique enables high-precision absolute positioning by utilizing a single GNSS receiver to receive satellite signals, while employing models, empirical formulas, and parameter estimation methods to handle delay errors in the propagation paths between the GNSS receiver and the satellites [14]. Assuming that the satellite s is observed by the GNSS receiver r, the raw pseudorange P and carrier phase L observation equations can be expressed as follows [14],[15]:

$$P^s_{r,j} = \rho^s_r + dt_r - dt^s + T^s_r + \gamma_j \cdot I^s_{r,i} + d_{r,j} - d^s_j + \varepsilon^s_{r,j} \tag{1}$$

$$L^s_{r,j} = \rho^s_r + dt_r - dt^s + T^s_r - \gamma_j \cdot I^s_{r,i} + \lambda^s_j (N^s_{r,j} + b_{r,j} - b^s_j) + \xi^s_{r,j} \tag{2}$$

$$\gamma_j = \frac{f_i^2}{f_j^2} \tag{3}$$

where: f represents the carrier signal frequency, with subscripts i and j denoting the frequency indices; ρ^s_r refers to the receiver-satellite geometric distance; dt_r and dt^s represent the receiver and satellite clock offsets, respectively, in meters; T^s_r refers to the slant tropospheric delay; the ionospheric delay $I^s_{r,j}$ can be decomposed into a frequency-dependent scaling factor γ_j and an ionospheric delay $I^s_{r,i}$ at the frequency i; ε and ξ represent the measurement noise; d and b represent the uncalibrated code delay (UCD) and uncalibrated phase delay (UPD), respectively, which are corrected using code bias and phase bias products; and λ and N represent the wavelength of the carrier phase observation and its integer ambiguity, respectively. The relativistic effect, Sagnac effect, antenna windup effect, and antenna phase center offset can be corrected using models and empirical formulas [14].

Dual-Frequency Satellite-based PPP Model

These satellite-based PPP services provide precise satellite products, including orbit, clock offset, code bias, and phase bias products. The broadcast framework of these services is illustrated in Fig. 1. Satellites transmit ephemeris data and corresponding corrections to deliver precise satellite orbit and clock offset products. The code bias and phase bias products are used to correct the UCD and UPD, respectively [16].

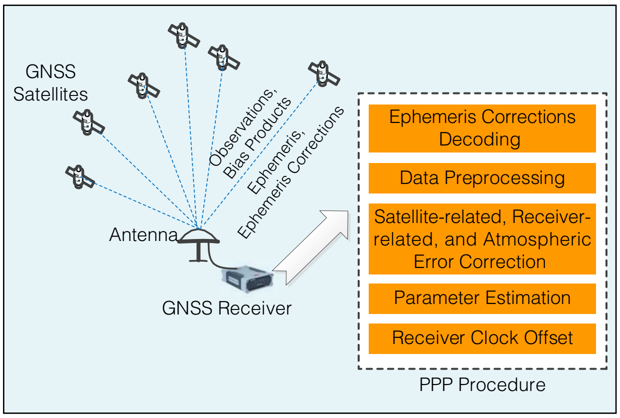

Fig. 1. Principle of satellite-based PPP.

Currently, while code bias products are available across all three services, the PPP-B2b and HAS services do not provide accessible phase bias products. The resulting differences in modeling will be highlighted in this section. Users primarily focus on recovering precise products, correcting various errors, and estimating unknown parameters.

Considering the frequency-dependent nature of ionospheric delay, the dual-frequency (DF) ionospheric-free (IF) PPP model, which eliminates the ionospheric delay effects, after applying the satellite product corrections, can be written as follows:

$$\begin{aligned} P_{r,ij}^{s} &= \alpha_{ij} \cdot \tilde{P}_{r,i}^{s} + \beta_{ij} \cdot \tilde{P}_{r,j}^{s} \\ &= \rho_r^s + dt_{r,ij} + T_r^s + \varepsilon_{r,ij}^{s} \end{aligned} \tag{4}$$

$$\begin{aligned} L_{r,ij}^{s} &= \alpha_{ij} \cdot L_{r,i}^{s} + \beta_{ij} \cdot L_{r,j}^{s} \\ &= \rho_r^s + dt_{r,ij} + T_r^s + \bar{\lambda}_{ij}^{s} \bar{N}_{r,ij}^{s} + \xi_{r,ij}^{s} \end{aligned} \tag{5}$$

$$\begin{cases} \alpha_{ij} = \dfrac{f_i^2}{f_i^2 - f_j^2}, \beta_{ij} = -\dfrac{f_j^2}{f_i^2 - f_j^2} \\ dt_{r,ij} = dt_r + (\alpha_{ij} \cdot d_{r,i} + \beta_{ij} \cdot d_{r,j}) \\ \lambda_{ij}^{s} N_{r,ij}^{s} = \alpha_{ij} \cdot \lambda_i^s (N_{r,i}^{s} + b_{r,i} - b_i^s) + \beta_{ij} \cdot \lambda_j^s (N_{r,j}^{s} + b_{r,j} - b_j^s) \\ \bar{\lambda}_{ij}^{s} \bar{N}_{r,ij}^{s} = \lambda_{ij}^{s} N_{r,ij}^{s} - (\alpha_{ij} \cdot d_{r,i} + \beta_{ij} \cdot d_{r,j}) \\ \rho_r^s = \tilde{\rho}_r^s + \boldsymbol{\mu}_r^s \cdot \Delta \mathrm{x} \end{cases} \tag{6}$$

where: $P_{r,ij}^{s}$ and $L_{r,ij}^{s}$ represent the IF pseudorange and carrier phase observations, respectively; $\tilde{P}$ represents the pseudorange observation corrected by code bias products; α and β are the IF coefficients; $dt_{r,ij}$ represents IF receiver clock offset to be estimated in the PPP solution; ρ_r^s can be expressed as a linear combination of the calculated geometric distance $\tilde{\rho}_r^s$, the receiver coordinates vector increments $\Delta\boldsymbol{x}$, and the unit satellite-receiver vector $\boldsymbol{\mu}_r^s$; $\lambda_{ij}^{s} N_{r,ij}^{s}$ represents the ambiguity resulting from the DF IF combination; and $\bar{\lambda}_{ij}^{s} \bar{N}_{r,ij}^{s}$ represents the float ambiguity to be estimated. $\bar{\lambda}_{ij}^{s} \bar{N}_{r,ij}^{s}$ encompasses both the combined ambiguity term $\lambda_{ij}^{s} N_{r,ij}^{s}$ and the receiver-related hardware delay $(\alpha_{ij} \cdot d_{r,i} + \beta_{ij} \cdot d_{r,j})$ which is associated with the estimated receiver clock offset $dt_{r,ij}$. When phase bias products are available, the estimated ambiguity can be resolved as an integer ambiguity [17]. In the absence of phase bias products, it remains estimated as a float ambiguity. Thus, the estimated parameter vector $\bar{X}$ can be written as:

$$\bar{X} = \begin{bmatrix} \Delta x & dt_{r,ij} & T_r^s & \bar{\lambda}\bar{N} \end{bmatrix} \tag{7}$$

where $\bar{\lambda}\bar{N}$ represents the estimated ambiguity vector, composed of $\bar{\lambda}_{ij}^{s} \bar{N}_{r,ij}^{s}$, whose dimension is determined by the number of observed satellites. Consequently, a minimum of five satellites with precise products is required to obtain a PPP solution [11]. In this study, ambiguities were estimated as float values. The station coordinates were estimated as constants. The tropospheric dry delay was corrected using the Saastamoinen model [18] and the GMF model [19]. The zenith wet delay was estimated as a random walk process. The receiver clock offset $dt_{r,ij}$ was estimated as white noise.

PPP Time Transfer

The receiver clock offset $dt_{r,ij}$ estimated by PPP is the deviation of the local time t_{loc} from the product reference time t_{ref}. By differencing the estimated receiver clock offsets between two stations within the time link, the reference time can be eliminated, enabling a direct differentiation of local time, expressed as:

$$\Delta t = dt_{\mathrm{A},ij} - dt_{\mathrm{B},ij} = (t_{\mathrm{loc,A}} - t_{\mathrm{ref}}) - (t_{\mathrm{loc,B}} - t_{\mathrm{ref}}) = t_{\mathrm{loc,A}} - t_{\mathrm{loc,B}} \tag{8}$$

where Δt represents the time comparison results for the time link.

Taking the post-processed PPP time link as the reference, the error sequence of real-time PPP time comparisons is obtained. The standard deviation (STD) of these errors is used to quantify the precision of the time comparison, and is expressed as:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2} \tag{9}$$

where x_i represents the sequence of time comparison errors; n represents the number of epochs in the error sequence; and $\bar{x}$ is the mean value of the sequence.

Comparison of Broadcast Frameworks

Currently, all three services provide precise satellite orbits, clock offset, and code bias products, while the MADOCA-PPP service additionally provides phase bias and ionospheric corrections. All products are provided in the form of state space representation (SSR) [20] correction. Orbit and clock offset corrections refine the ephemeris to recover precise satellite orbit and clock offset, while code bias and phase bias are employed to correct hardware delays in pseudorange and carrier phase observations.

However, notable differences exist in the encoding of satellite products into real-time binary data streams and their broadcasting frameworks. The encoding and broadcasting of PPP-B2b, HAS, and MADOCA-PPP are illustrated in Fig. 2, Fig. 3, and Fig. 4, respectively. The PPP-B2b products are

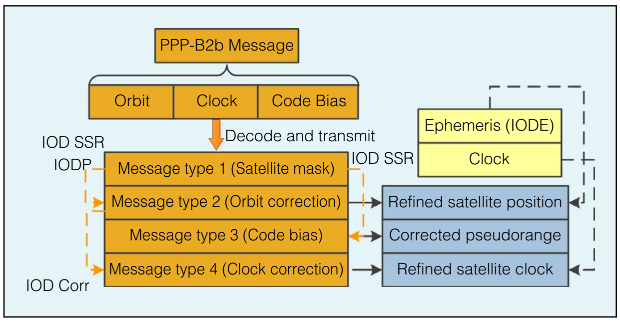

Fig. 2. PPP-B2b message structure.

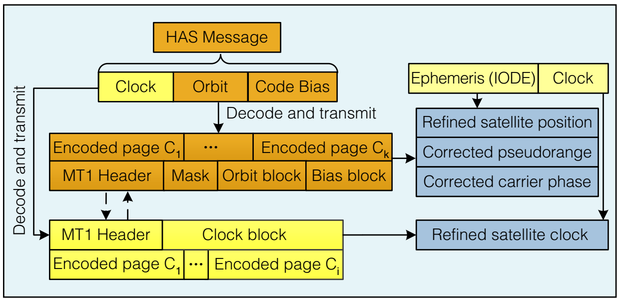

Fig. 3. HAS message structure.

decoded into four message types, with compatibility maintained through the issue of data (IOD). In contrast, the HAS products encode message types (MT) 1 to 3 as MT1-slow, while clock offset corrections are encoded separately as MT1-fast in Section 7.6 of its interface control document (ICD) [9]. The MADOCA-PPP service encodes orbit, clock offset, and bias products into six message types for broadcasting via L6E, while ionospheric corrections are transmitted through L6D.

In contrast to the PPP-B2b service, which encodes message types categorically, the MADOCA-PPP service transmits these encoded message types at fixed pattern, delivering a complete set of products every 30 seconds in Section 4.1 of its ICD [10]. This fixed transmission pattern simplifies message matching for users, as it only requires confirming whether the continuously decoded products belong to the same data set. Thus, the MADOCA-PPP service reduces the complexity of message matching for users. While the PPP-B2b service minimizes message length, it enhances transmission efficiency, facilitating the inclusion of error correction coding.

In terms of service broadcasting, the PPP-B2b service transmits corrections via three geostationary earth orbit (GEO) satellites, operating as a regional PPP

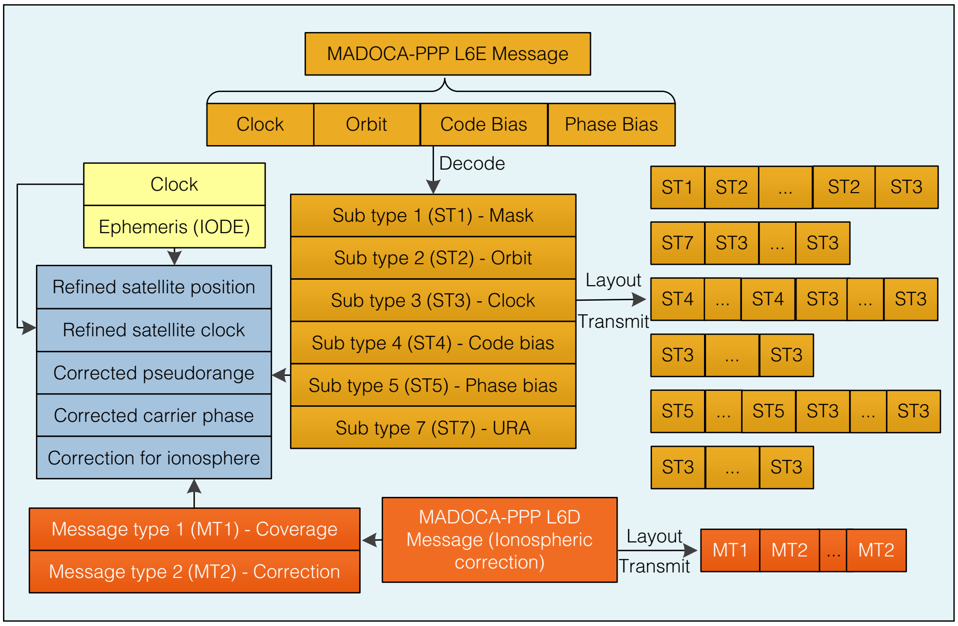

Fig. 4. MADOCA-PPP message structure.

Table 1 – Detailed differences in the PPP-B2b, HAS, and MADOCA-PPP services			
Item	**BDS PPP-B2b**	**Galileo HAS**	**QZSS MADOCA-PPP**
Currently supported systems	GPS, BDS	GPS, Galileo	GPS, GLONASS, Galileo, QZSS
Currently supported corrections	Satellite orbit, clock, code bias	Satellite orbit, clock, code bias	Satellite orbit, clock, code bias, phase bias, ionospheric correction
Broadcast channel	Satellites	Satellites and Internet (RTCM)	Satellites and Internet (Compact SSR)
Carrier satellites	BDS GEO satellites (C59, C60, C61)	All operational Galileo satellites	All operational QZSS satellites
Carrier signal	BDS B2b (1207.14 MHz)	Galileo E6-B (1278.75 MHz)	QZSS L6E and L6D (1278.75 MHz)
Corrected ephemeris	BDS CNAV1, GPS LNAV	Galileo I/NAV, GPS LNAV	GPS LNAV, GLONASS GLONASS-K(M), Galileo I/NAV, QZSS LNAV
Time system	BeiDou Navigation Satellite System Time (BDT)	Galileo System Time (GST)	QZSS Time (QZSST)

service since product estimates primarily rely on China's tracking stations. Moreover, the binary data stream for the PPP-B2b service broadcasted by the GEO satellites remains consistent. The MADOCA-PPP estimates its products using a global observation network [12] and broadcasts them through QZSS satellites. Additionally, its products are also disseminated via the Internet. In contrast to the PPP-B2b service, the MADOCA-PPP service features a longer message length and does not permit switching between QZSS satellites within the 30-second reception period for a set of satellite products. Conversely, the HAS service provides global coverage due to its distinct broadcasting scheme.

To enhance reception efficiency, the Galileo HAS employs the high-parity vertical reed-solomon (HPVRS) encoding scheme, fragmenting a set of products through an encoding matrix and distributing the fragments across different Galileo satellites [21]. Users need to receive a sufficient number of the fragments, which can come from the same or different Galileo satellites, and these are restored into HAS products via HPVRS decoding to optimize user reception. Additionally, the complex decoding process for HAS products is supported by C/C++ programming packages [22]. Table 1 summarizes the detailed differences in broadcasting among the three services. Users may preselect a suitable service by considering the specific features of each broadcast framework, the complexity of decoding, and the associated advantages and drawbacks.

Service Coverage and Availability

Observations from 180 globally distributed stations, sampled at 30-second intervals from day of year (DoY) 278 to 282 in 2024, were used to analyze the PPP coverage and availability supported by the three services. The distribution of these stations is illustrated in Fig. 5, where 45 stations are specifically selected for analyzing the regional PPP-B2b service, represented by red and green dots. The PPP-B2b and HAS products were cashed using Septentrio Mosaic-X5 and SINO K803 receivers, each equipped with their respective receiving modules, while the MADOCA-PPP data were obtained from the official online archive (https://sys.qzss.go.jp/dod/en/archives.html). The analysis of PPP service coverage was conducted by evaluating the average number of satellites with corrections. A minimum threshold of five satellites with corrections was set to support PPP solutions. The proportion of epochs with sufficient corrections relative to the total number of epochs was considered as an indicator of service availability.

The interpolation results for the average number of satellites with corrections and PPP availability for the PPP-B2b, HAS, and MADOCA-PPP services are presented in Fig. 6, Fig. 7 and Fig. 8, respectively. The statistical results for the PPP-B2b service (20° N~55° N, 75° E ~ 150° E), HAS, and MADOCA-PPP service (60° S~60° N, 70° E ~ 160° W) are summarized in Table 2. Due to the limitations of the observation network generating the products, PPP-B2b availability is primarily concentrated in China and its surrounding regions, with BDS service availability reaching up to 99.93%. In contrast, the Galileo HAS offers global coverage, providing an average of 8.55 GPS satellites

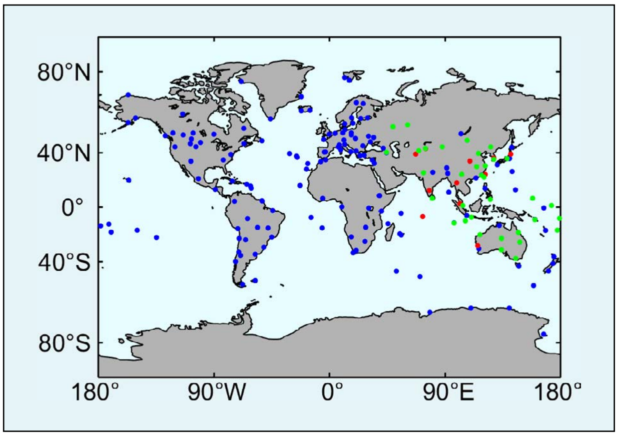

Fig. 5. Geographical distribution of stations. Blue dots represent stations evaluated for the HAS and MADOCA services, green dots for stations assessed across all three services, and red dots for stations evaluated for the PPP-B2b service.

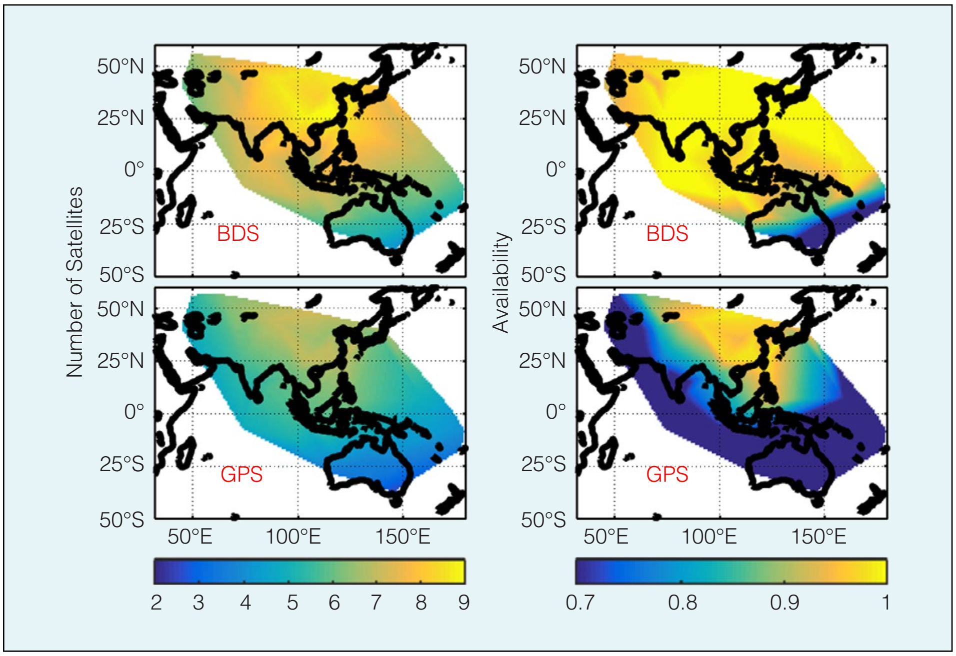

Fig. 6. Average number of satellites with corrections and PPP availability of the PPP-B2b service.

and 8.44 Galileo satellites with corrections contributing to PPP calculations at each epoch on the user side. This results in PPP availability rates of 98.84% and 99.17%, respectively. The MADOCA-PPP provides services for the Asia-Pacific region through QZSS satellites, with an average of 10.06 corrected GPS satellites and an availability rate of 99.99%. However, the number of corrected GLONASS satellites is relatively low, while the corrected Galileo satellites and their availability are comparable to those of HAS. In summary, the real-time products of all three services exhibit good continuity and provide precise products for a sufficient number of satellites, meeting the requirements of real-time PPP. Relative to Fig. 6, Fig. 7, and Fig. 8, high-availability services could be selected at both user stations to enable continuous, stable, and reliable real-time PPP time transfer.

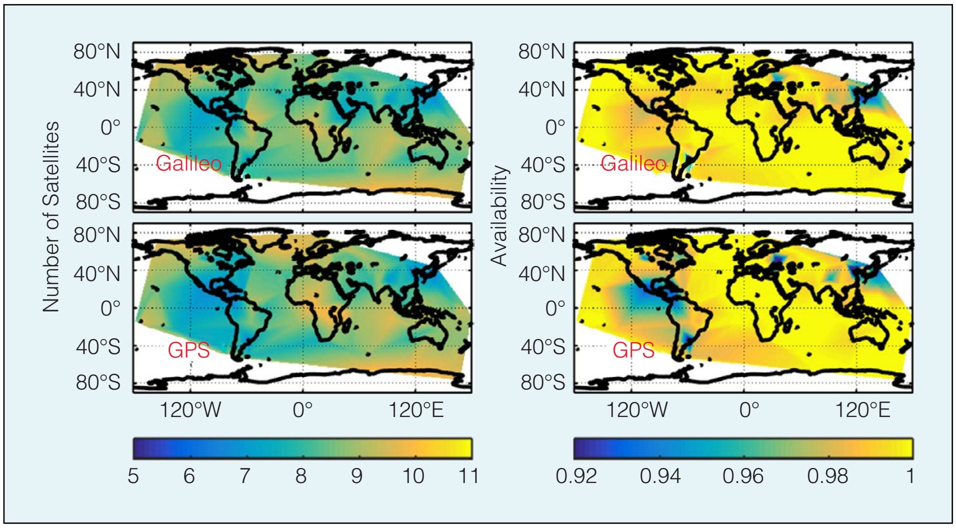

Fig. 7. Average number of satellites with corrections and PPP availability of the HAS service.

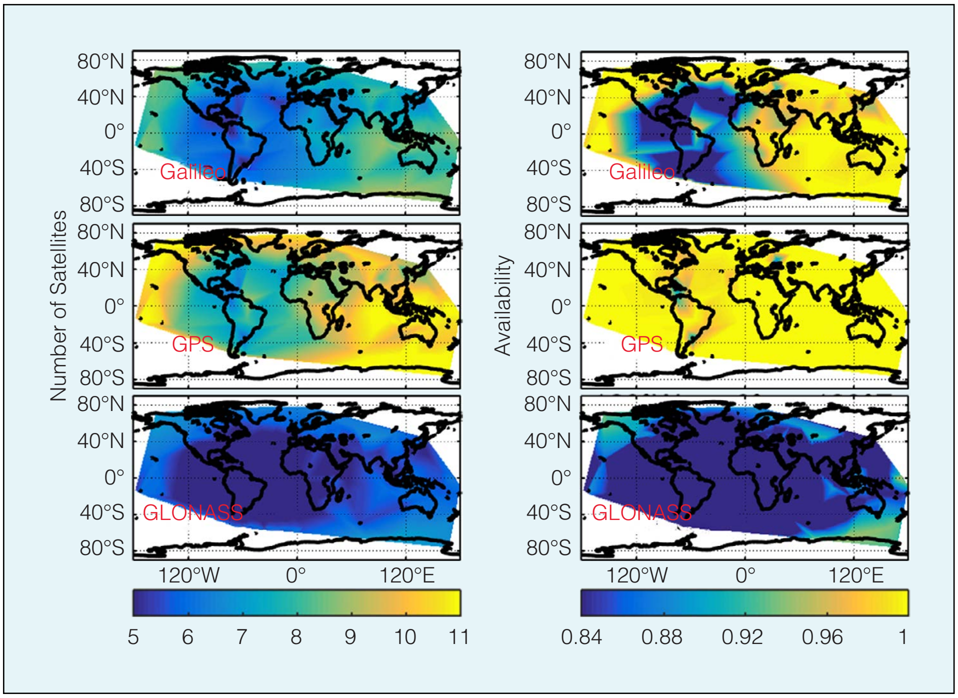

Fig. 8. Average number of satellites with corrections and PPP availability of the MADOCA-PPP service.

Time Transfer Results

The results sequences of applying the three services to PPP time comparison are presented and analyzed. To validate the accuracy of PPP time transfer across the three services, it is essential to account for potential satellite shortages that may cause anomalies in estimated receiver clock offset. Due to the varying PPP availability of the three services, ensuring enough satellites is critical for accurately assessing time transfer accuracy. In this experiment, PPP-B2b time transfer utilizes both the GPS and BDS systems, while HAS and MADOCA-PPP time transfers are based

Table 2 – Average number of satellites with corrections and service availability

Item	BDS PPP-B2b		Galileo HAS		QZSS MADOCA-PPP		
	GPS	BDS	GPS	Galileo	GPS	Galileo	GLONASS
Average number of satellites	6.79	8.53	8.55	8.44	10.06	7.79	5.77
Service availability	96.47%	99.93%	98.84%	99.17%	99.99%	98.90%	81.16%

on the GPS and Galileo systems. To compare the time transfer accuracy of the three services, three user stations were selected within the overlapping coverage area of the high-availability services provided by all three. The National Time Service Center (NTSC), a TAI time-keeping laboratory, serves as the central node, linking IGS stations (ULAB and USUD) to form time links for time comparison accuracy analysis. The geographic distribution of the stations and the baseline lengths of the two time links are shown in Fig. 9.

With reference to GeoForschungZentrum (GFZ) post-processed precise products GBM, the time comparison errors for the NTSC-ULAB and NTSC-USUD time links, based on real-time products, are presented in Fig. 10 and Fig. 11, with baseline lengths of 1506 km and 2631 km, respectively. Even with the longer baseline of 2631 km, the time comparison errors across all three services fluctuated within a narrow range of ±0.5 ns. The STD values of the time comparison errors are summarized in Table 3. When stable and continuous PPP time transfer is performed, the comparison accuracy is affected by the baseline length. The MADOCA-PPP time transfer results exhibit smaller error fluctuations, while the time comparison accuracy using the PPP-B2b and HAS products is comparable. This can be attributed to the more precise satellite clock offset products provided by the MADOCA-PPP service [12]. For the NTSC-ULAB time link, the STD value of the time comparison error for MADOCA-PPP was 78 ps, compared to 121 ps for PPP-B2b and 148 ps for HAS. Therefore, provided that high availability is guaranteed for all three services, they can support real-time continuous, stable, and reliable PPP time comparisons at the sub-nanosecond level, with time comparison errors remaining within ±0.5 ns, even over a baseline length of up to 2631 km.

It should be emphasized that the selection of the three time stations was exclusively for the purpose of comparing the time comparison accuracy of the three real-time PPP services. The user stations involved in time transfer may be located anywhere. Furthermore, the feasibility of Galileo HAS for global stable time transfer has been demonstrated, with baseline lengths extending up to 12295 km and time transfer errors fluctuating within ±1 ns. Considering that both MADOCA-PPP and PPP-B2b are regional services, the status and characteristics of the real-time PPP services supported in different regions are different. To facilitate continuous and stable PPP time transfer at the sub-nanosecond level, time users are advised to select suitable high-continuity real-time services with reference to Fig. 6, Fig. 7 and Fig. 8, taking into careful consideration the broadcast framework architecture, decoding complexity, as well as the respective advantages and limitations of each service.

Conclusions

The PPP technique has been used in TAI time link calculations for decades. Recently, the dissemination of real-time precise orbit and clock offset products via satellites has greatly enhanced both research and implementation of real-time PPP, reducing dependence on network connectivity. However,

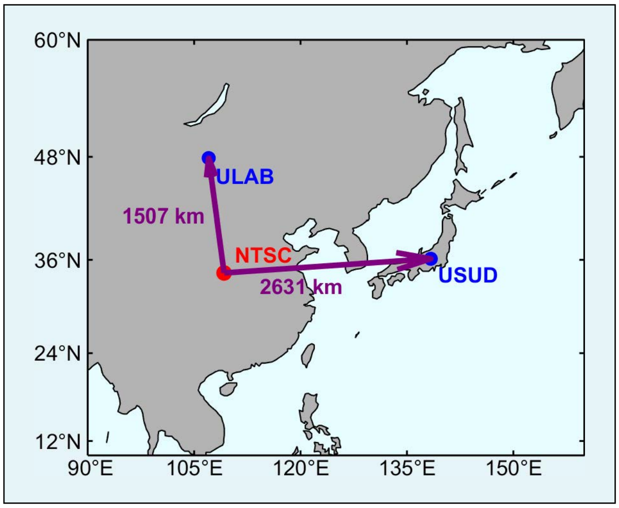

Fig. 9. Geographical distribution of stations. The NTSC (red) at the center connects other stations (blue) to form time comparison link.

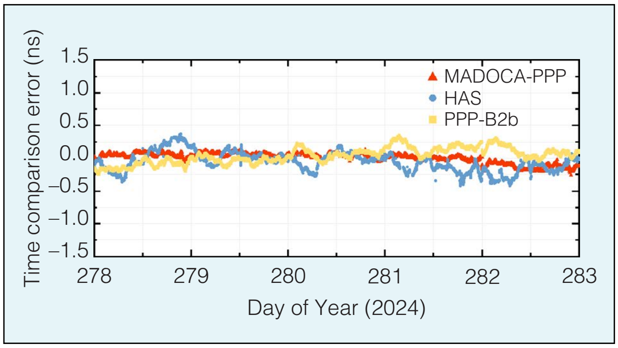

Fig. 10. Time comparison error series for time link NTSC-ULAB from DoY 278 to 282 in 2024.

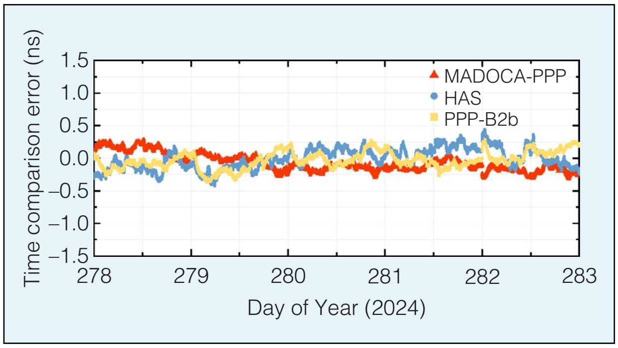

Fig. 11. Time comparison error series for time link NTSC-USUD from DoY 278 to 282 in 2024.

Table 3 – STD values of time comparison error (unit: ps)

Time link \ Model	PPP-B2b	HAS	MADOCA-PPP
NTSC-ULAB	121	148	78
NTSC-USUD	133	159	111

due to bandwidth limitations, satellite broadcasting requires precise products to be segmented and encoded for transmission, necessitating users to recover the complete product on their end. This study conducts a comprehensive comparison of the broadcast encoding, service coverage, availability, and time transfer results for BDS PPP-B2b, Galileo HAS, and QZSS MADOCA-PPP.

BDS provides regional PPP services to China and its surrounding areas via three GEO satellites, while Galileo offers global PPP services via its operational satellites through the HPVRS scheme. Additionally, QZSS supports regional PPP services across the Asia-Pacific region through its satellite constellation. Both PPP-B2b and MADOCA-PPP segment their products by message types; MADOCA-PPP employs a fixed transmission pattern, simplifying message matching process for users. PPP-B2b, on the other hand, shortens message length and incorporates error correction to ensure transmission accuracy. To improve user reception efficiency, HAS adopts the HPVRS encoding scheme, which adds complexity to product reception and recovery for users.

Regarding PPP service coverage and availability, PPP-B2b achieves a PPP availability of 99.93% for BDS users within the region (20° N to 55° N, 75° E to 150° E). Conversely, HAS demonstrates comparable global coverage and service availability for GPS and Galileo users, with an average of approximately 8.5 satellites and nearly 99% availability. MADOCA-PPP, covering the region between 60° S and 60° N and 70° E to 160° W, reports an average of 10.06 corrected GPS satellites and 99.99% availability. The real-time products of all three services demonstrate adequate continuity and offer precise products for a sufficient number of satellites, thereby satisfying the requirements of real-time PPP. In the future, a software solution will be developed to enable simultaneous decoding of HAS, PPP-B2b, and MADOCA-PPP products for real-time PPP time transfer. As these products continue to evolve, the application of PPP with ambiguity resolution technique to real-time time transfer will also be explored and integrated.

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